

Richard Álvarez

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Research Interests

Human-centered artificial intelligence; human-computer interaction; reliable tool-using AI systems; computational film and media; multimodal creative tools; and AI-assisted media production.

Education

Kenyon College

Aug 2020–May 2024

Bachelor of Arts in Film

Gambier, Ohio

- Minor in History; concentration in the Integrated Program in Humane Studies.
- Degree awarded May 18, 2024.

Publications and Posters

- **Álvarez, Richard Anthony.** *A Retrieval-Augmented Film Recommendation System*. Poster, *IPHS 484: Senior Seminar*, Paper 34, Digital Kenyon, 2024. digital.kenyon.edu/dh_iphs_ss/34
- **Álvarez, Richard Anthony.** *Captivating Video Editing Secrets: Unsupervised Deep Learning and PySceneDetect Analysis of Creative Strategies*. *IPHS 484: Senior Seminar*, Paper 22, Digital Kenyon, 2023. digital.kenyon.edu/dh_iphs_ss/22

Professional Experience

AI Solutions Engineer

Sep 2025–Present

Layer 7 Systems, Independent Contractor

Remote

- Built and operated a stateful TypeScript, Bun, and Express MCP service connecting Retell voice agents with Autotask and Microsoft Graph/Teams.
- Implemented identity and company resolution, contact-safe ticket access, authorization gates, ticket actions, technician routing, post-call processing, and human handoff.
- Developed an authenticated React operations dashboard with structured logging, health checks, alerts, cost tracking, and operator controls.
- Maintained more than 750 regression tests; a September 2026 verification run passed 756 tests across 84 files with zero failures.
- Deployed Docker services through GitHub Actions and investigated duplicate tickets, stale identity state, API drift, retry failures, transfer defects, and VPS incidents.
- Analyzed 60 workflow recordings from 14 users and identified 176 moments suitable for AI assistance.
- Built a Next.js and Neon/PostgreSQL learning platform for 34 trainees across four courses and 16 modules.

Freelance Software and Web Developer

Oct 2024–Present

Richard Álvarez, Sole Proprietorship

Chicago, Illinois

- Design and deploy responsive applications using Next.js, React, Node.js, Tailwind CSS, Stripe, PostgreSQL, authentication, and object storage.
- Translate client requirements into scoped implementations, deployment plans, documentation, training, hosting, and ongoing content workflows.
- Maintain client applications after launch and respond to deployment, security, performance, and content-management requirements.

IT Assistant

Sep 2023–Feb 2024

*Kenyon College Library and Information Services**Gambier, Ohio*

- Supported workstation imaging, hardware deployment, printer and audiovisual troubleshooting, secure-drive destruction, and classroom technology maintenance.
- Created a spreadsheet-based scheduling tool for recurring classroom maintenance.

Selected Research and Software Projects**Vigil**

2025–2026

- Built an open-source, self-hosted email intelligence agent with multi-model routing, encrypted BYOK credentials, persistent memory, alerts, audit logs, and per-invocation cost tracking.
- TypeScript, Bun, Hono, SQLite FTS5, Next.js, and Cloudflare.

Retrieval-Augmented Film Recommendation System

2024

- Generated user-specific retrieval queries, enriched film metadata through external APIs, and produced recommendations grounded in retrieved context.
- Node.js, LangChain, OpenAI, OMDb, and TMDb.

Computational Video-Editing Analysis

2023

- Analyzed editing patterns in short-form online video using automated scene detection and unsupervised analysis.
- Python, PySceneDetect, and computational video analysis.

Half-Time Draw Predictor

2026

- Built a sports-analytics pipeline combining market odds with team fundamentals to identify potentially mispriced half-time draw bets.
- Python, scikit-learn, XGBoost, and Jupyter.

Chroma

2026

- Designed a C++20 terminal roguelike deckbuilder in which musical relationships and color form the underlying mechanical system.
- C++20, ncurses, procedural audio, and terminal-interface design.

GREasyVocab

2024

- Built an AI-assisted vocabulary application with authentication, persistent user data, and contextual flashcard generation.
- TypeScript, Next.js, PostgreSQL, and OpenAI APIs.

Bitset Game of Life

2024–2026

- Implemented Conway's Game of Life in C++ with a bitset-oriented grid representation.

Kubrick-Style Script Generation Pipeline

2024

- Experimented with few-shot and chain-of-thought prompting in a multi-stage JavaScript pipeline for film-script generation.

Rotten Tomatoes Review Sentiment Analysis

2024

- Collected and analyzed user reviews with Python, BeautifulSoup, NLTK, and VADER to compare sentiment with platform freshness labels and audience ratings.

DV Tools

2025

- Created Shell and FFmpeg utilities for ingesting and processing footage from legacy DV/DVI camera workflows.

Selected Applied AI Systems

Business and AI Training Platform 2026

- Built and deployed a Next.js, TypeScript, and Neon/PostgreSQL learning platform serving 34 trainees.

Workflow Capture and Analysis Pipeline 2026

- Built an Electron, TypeScript, FFmpeg, and Gemini workflow-capture and AI-assisted analysis pipeline.

Configurable Voice-Reception Template 2025

- Built a reusable JavaScript, Retell, and n8n voice-intake template with configurable business context, transfer behavior, webhooks, and integration boundaries.

Message-Box Auto-Attendant Template 2025

- Built caller-message capture, spam screening, business-hours transfer, and structured notifications.

VibeVoice Evaluation Prototype 2025

- Evaluated low-latency self-hosted speech synthesis on a single GPU using Python, FastAPI, WebSockets, and VibeVoice.

Selected Professional Projects

Joaquin Morales Cinematography Portfolio 2024–2026

- Designed and deployed a cinematography portfolio with responsive media presentation, object storage, authentication, and client-managed content.

Machine Television 2024–2025

- Built a Next.js commercial website and Stripe payment flow for an independent creative business.

BDPA HSCC Website Prototype 2025

- Created initial Next.js and TypeScript templates for a public-facing competition website; the repository contains an early prototype rather than a production deployment.

Personal Portfolio 2024–Present

- Maintains a public record of software, AI systems, web work, research, and creative practice.

Professional and Community Service

High School Computer Competition Judge 2024–2025

BDPA Chicago, Illinois

- Reviewed and scored student programming submissions for the 2024 and 2025 national competitions and completed formal judge training.

Professional Certification

CompTIA IT Fundamentals (ITF+) 2024

Technical Skills

- **Languages and application development:** TypeScript, Python, JavaScript, SQL, C++, HTML/CSS, Shell, Bun, Node.js, Express, Hono, FastAPI, React, Next.js, Electron, and Zod.
- **Data and applied AI:** PostgreSQL, Neon, SQLite FTS5, MCP, REST APIs, tool calling, structured outputs, retrieval-augmented generation, context engineering, OpenAI, Anthropic, Gemini, and Retell.
- **Integration and operations:** Autotask, Microsoft Graph/Teams, Docker, GitHub Actions, Linux/VPS operations, CI/CD, automated testing, validation, structured logging, observability, and incident response.
- **Computational media:** PySceneDetect, FFmpeg, NLP and sentiment analysis, computational film analysis, and interactive media.